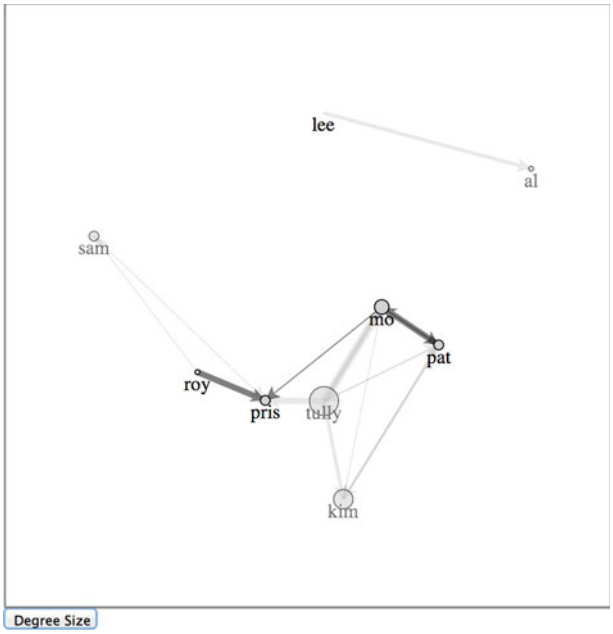



**Figure 6.18 The network has been filtered to only show nodes with more than 20 followers, after clicking the Degree Size button. Notice that Lee, with no connections, has a degree of 0 and so the associated circle has a radius of 0, rendering it invisible. This catches two processes in midstream, the transition of nodes from full to 0 opacity and the removal of edges.**

### 6.2.6 Removing and adding nodes and links

When dealing with networks, you may want to filter the networks or give the user the ability to add or remove nodes. To filter a network, you need to `stop()` it, remove any nodes and links that are no longer part of the network, rebind those arrays to the force layout, and then `start()` the layout.

This can be done as a filter on the array that makes up your nodes. For instance, we may want to only see the network of people with more than 20 followers, because we want to see how the most influential people are connected.

But that's not enough, because we would still have links in our layout that reference nodes that no longer exist. We'll need a more involved filter for our links array. By using the `.indexOf` function of an array, though, we can easily create our filtered links by checking to see if the source and target are both in our filtered nodes array. Because we used key values when we first bound our arrays to our selection in listing 6.8, we can use the `selection.exit()` behavior to easily update our network. You can see how to do this in the following listing and the effects in figure 6.18.

**Listing 6.9 Filtering a network**

```
function filterNetwork() {
      force.stop();
      var originalNodes = force.nodes();
      var originalLinks = force.links();
      var influentialNodes = originalNodes.filter(function (d) {
            return d.followers > 20;
      });
```

**Accesses the current array of nodes and array of links associated with the force layout**